



HK IMO Training 2023-2024

Problem Set 11 Suggested Solutions



Problem 41 (Argentina MO 2020 Problem 5 modified). Let n be a positive integer. Find the largest possible value of

$$S = x_1x_2x_3 + x_4x_5x_6 + \cdots + x_{3n-2}x_{3n-1}x_{3n} + x_{3n+1}$$

where $(x_1, x_2, \dots, x_{3n+1})$ is a permutation of $(1, 2, \dots, 3n+1)$.

Solution. The answer is $\frac{1}{4}(27n^4 + 54n^3 + 21n^2 - 6n + 4)$.

For any integers $a, b, c > 1$, we have $abc + 1 > ab + c$ by the rearrangement inequality (because $ab > 1$ and $c > 1$). Therefore, we may assume $x_{3n+1} = 1$ (otherwise the value of S is increased by swapping 1 with any other number). We say that the three numbers $x_{3j-2}, x_{3j-1}, x_{3j}$ belong to the same triple for each $j = 1, 2, \dots, n$.

Since there are finitely many permutations, we can choose a permutation which maximizes S . Consider any two triples (a, b, c) and (d, e, f) . We say that a term in a, b, c, d, e, f is *big* if it is one of the three largest terms among a, b, c, d, e, f . By the pigeonhole principle, at least two big terms belong to the same triple. WLOG assume a, b are two big terms. If c is not the remaining big term, we may assume d is a big term. Thus, we have $ab > ef$ and $d > c$. By the rearrangement inequality, we have $abd + cef > abc + def$. This shows the value of S is increased if we swap c and d , contradiction. Therefore, the big terms must belong to the same triple.

Using the above observation, we know that $3n+1$ and $3n$ cannot belong to different triples, and $3n-1$ must belong to the same triple as $3n+1$ and $3n$. This means the three largest terms belong to the same triple. Similarly, the next three largest terms belong to the same triple, etc. In other words, S is maximized when

$$\begin{aligned} S &= (3n+1)(3n)(3n-1) + (3n-2)(3n-3)(3n-4) + \cdots + (4)(3)(2) + 1 \\ &= \sum_{k=1}^n (3k+1)(3k)(3k-1) + 1 \\ &= \sum_{k=1}^n (27k^3 - 3k) + 1 \\ &= \frac{1}{4}(27n^4 + 54n^3 + 21n^2 - 6n + 4). \end{aligned}$$

□

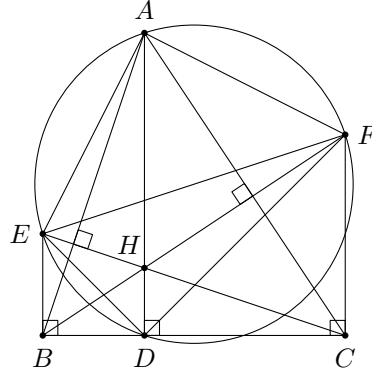
Problem 42 (Argentina MO 2020 Problem 6). Let $n \geq 3$. Two players X and Y play a game in a regular n -gon with a vertex A which is assigned as a trap. Initially, X places a token at one of the vertices other than A . In each round, Y announces a positive integer k (not necessarily the same each time). Then X moves the token by k vertices in the clockwise or anticlockwise direction. Find all n for which X has a strategy that guarantees the token never falls into the trap regardless of how Y plays.

Solution. Answer: n is not a power of 2.

If n is not a power of 2, let p be an odd prime dividing n . We label the vertices from 0 to $n-1$ modulo n such that vertex 0 is the trap. It suffices to show that X has a strategy which guarantees the token never lands on a vertex m with $p \mid m$. For instance, X can place the token at vertex 1 at the beginning. Suppose the token lands on a vertex m at some step with $p \nmid m$. When Y announces a positive integer k , one of $m+k$ and $m-k$ is not divisible by p , since p does not divide $(m+k) + (m-k) = 2m$. Thus, X can always choose a vertex which is not divisible by p as desired.

If n is a power of 2, we show that Y can guarantee the token falls into the trap. Suppose the token is placed at vertex m initially. Suppose Y announces m . Since $m-m=0$, X has to move the token to vertex $m+m=2m$. Next, suppose Y announces $2m$, then the token has to be moved to vertex $2m+2m=4m$. Inductively, the token has to be moved to vertex $2^j m$ after j steps. As n is a power of 2, it divides $2^j m$ for any sufficiently large j . This means the token is eventually forced to move to the trap. □

Problem 43 (Colombia MO 2020 Problem 5). In acute $\triangle ABC$, let D be the foot of altitude from A . Let E be a point on the altitude from C such that $BE \perp BC$, and let F be a point on the altitude from B such that $CF \perp BC$. Prove that $AD = BC$ if and only if A, E, D, F are concyclic.



Solution. We first provide a proof for the general case $AB \neq AC$. Let H be the orthocentre of $\triangle ABC$. Since $BE \parallel HD \parallel CF$, we have $\frac{BE}{CF} = \frac{EH}{CH} = \frac{BD}{CD}$. Together with $\angle EBD = 90^\circ = \angle FCD$, we deduce $\triangle BDE \sim \triangle CDF$. It follows that $\angle BDE = \angle CDF$. Therefore, $\angle ADE = 90^\circ - \angle BDE = 90^\circ - \angle CDF = \angle ADF$. Note that $DE \neq DF$ as $BD \neq CD$. Thus, A, E, D, F are concyclic iff $AE = AF$.

As $AB \perp CE$, $AC \perp BF$ and $AD \perp BC$, we have

$$\begin{aligned} AE^2 &= BE^2 - BC^2 + AC^2, \\ AF^2 &= CF^2 - BC^2 + AB^2, \\ AB^2 - AC^2 &= BD^2 - CD^2. \end{aligned}$$

Combining these, we see that $AE = AF$ iff $BE^2 - CF^2 = BD^2 - CD^2$, or $BE^2 - BD^2 = CF^2 - CD^2$. Recall that $\triangle BDE \sim \triangle CDF$. These triangles are not congruent since $BD \neq CD$ in this case. Therefore, $BE^2 - BD^2 = CF^2 - CD^2$ iff $BE^2 - BD^2 = CF^2 - CD^2 = 0$ iff $BE = BD$ and $CF = CD$.

It is easy to see that $\triangle ABD \sim \triangle CEB$ as they have the same interior angles. If $AD = BC$, then they are congruent triangles, and hence $BD = EB$. By symmetry, $CD = FC$. From above, A, E, D, F are concyclic. Conversely, if A, E, D, F are concyclic, then $BE = BD$, so $\triangle ABD \cong \triangle CEB$. This implies $AD = BC$.

We are left with the case $AB = AC$. In this case, we have $\angle AED = \angle AFD$ by symmetry. Therefore, A, E, D, F are concyclic iff $\angle AED = \angle AFD = 90^\circ$, or just $\angle AED = 90^\circ$ by symmetry. Let $BD = CD = x$ and $AD = 2y$. Then $AD = BC$ iff $x = y$. Using Pythagoras' theorem and $\triangle ABD \sim \triangle CEB$, we deduce

$$\begin{aligned} AB^2 &= AC^2 = x^2 + 4y^2, \\ BE &= BD \times \frac{BC}{AD} = \frac{x^2}{y}, \\ DE^2 &= x^2 + \frac{x^4}{y^2}, \\ AE^2 &= BE^2 - BC^2 + AC^2 = \frac{x^4}{y^2} - 4x^2 + (x^2 + 4y^2). \end{aligned}$$

Thus, $\angle AED = 90^\circ$ iff $AE^2 + DE^2 = AD^2$ iff $\frac{2x^4}{y^2} - 2x^2 + 4y^2 = 4y^2$, or $2x^4 = 2x^2y^2$. Clearly, this holds iff $x = y$ as desired. $\square$

Problem 44 (Korea MO 2019 Problem 7). Let p be a prime number such that $p \equiv 1 \pmod{7}$. Prove that there exists a positive integer n such that $p \mid n^3 + n^2 - 2n - 1$.

Solution. Let g be a primitive root modulo p , and let $t = g^{\frac{p-1}{7}}$. Let n be any positive integer such that $n \equiv t + t^{-1} \pmod{p}$. We claim that $p \mid n^3 + n^2 - 2n - 1$. Indeed, this follows from

$$\begin{aligned} n^3 + n^2 - 2n - 1 &\equiv (t + t^{-1})^3 + (t + t^{-1})^2 - 2(t + t^{-1}) - 1 \\ &\equiv t^3 + t^2 + t + 1 + t^{-1} + t^{-2} + t^{-3} \\ &\equiv t^{-3}(t^6 + t^5 + \cdots + 1) \\ &\equiv t^{-3}(t^7 - 1)(t - 1)^{-1} \\ &\equiv t^{-3}(g^{p-1} - 1)(t - 1)^{-1} \\ &\equiv 0 \pmod{p}. \end{aligned}$$

(Note that $t - 1 \not\equiv 0 \pmod{p}$ and $g^{p-1} \equiv 1 \pmod{p}$.)

□